

**EE1101: Circuits and Network Analysis - End Exam**

Name: \_\_\_\_\_

Roll No: \_\_\_\_\_

**Instructions**

1. The exam lasts for 90 minutes, beginning at 10:00. Please ensure you submit your paper by 11:30.
2. Write your solutions in the answer book. Answers written elsewhere will not be considered for grading.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

Signature: \_\_\_\_\_

1. (8 points) Consider the circuit shown in Fig. 1.
  - (a) (2 points) Compute the Thevenin voltage across the port  $a-o$ .
  - (b) (2 points) Compute the Thevenin impedance as seen from the port  $a-o$ .
  - (c) (2 points) Compute the value of the resistance (say  $R_l$ ) that needs to be connected across the port  $a-o$  to extract maximum power from the source.
  - (d) (2 points) Compute the voltage across  $R_l$  under the condition of maximum power.

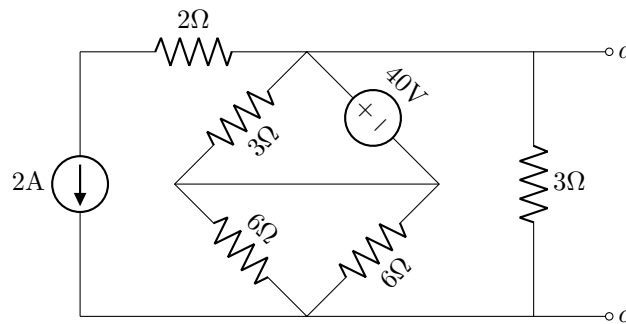


Fig. 1: Circuit for question 1

2. (4 points) Compute the  $h$ -parameters of the two port network shown in Fig. 2(a). (Clearly indicate if any of the parameters are not defined).
3. (6 points) Consider a two port network with impedance parameters (at a frequency  $\omega$ )  $\mathbf{z}_{11} = j5\omega$ ,  $\mathbf{z}_{22} = (1 - j\omega)$  and  $\mathbf{z}_{12} = \mathbf{z}_{21} = (-j\omega)\Omega$ . A voltage source with variable frequency  $\mathbf{V}_s$  is connected at port 1 and a resistance of  $R_l = 1\Omega$  is connected across port 2.
  - (a) (2 points) Compute the transfer function  $\mathbf{H}(j\omega) = \frac{\mathbf{V}_2}{\mathbf{V}_s}$  where  $\mathbf{V}_2$  is the voltage across the  $R_l$ .
  - (b) (4 points) Plot the approximate frequency response (Bode plot) of  $\mathbf{H}(j\omega)$ .

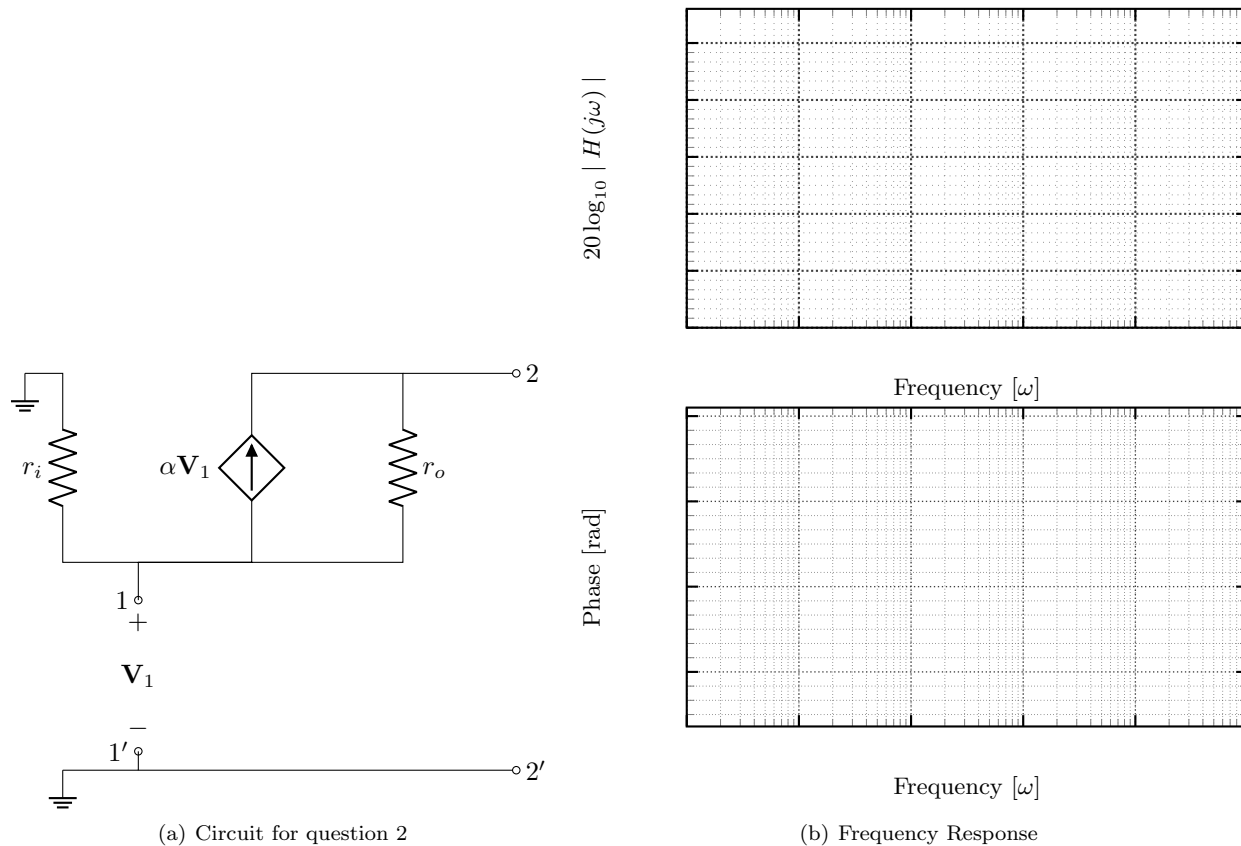


Fig. 2: Circuit for question 2 and space for frequency response

4. (6 points) An inductor is designed of the form shown in Fig. 3. The area of cross section is  $3 \text{ cm}^2$  and the mean cross length of the core is  $15 \text{ cm}$ . Assuming the air gap length to be  $1 \text{ mm}$  and the permeability of the core to be  $\frac{500}{\pi} \mu_0$ , compute the number of turns to achieve an inductance of  $6.0 \text{ mH}$  (Assume a 5% increase in the effective area of the air gap due to fringing effects and neglect leakage). Also compute the value of current that causes the core to saturate, assuming saturation flux density to be  $1.0 \text{ T}$ .

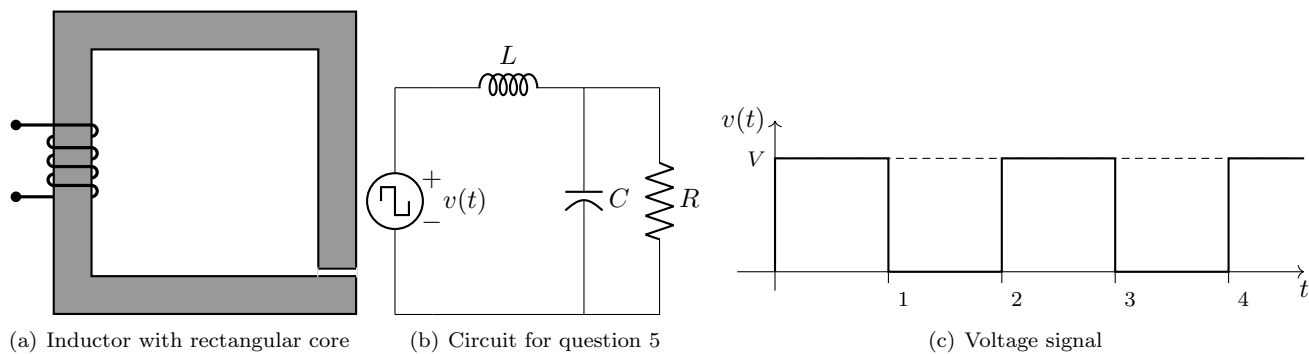


Fig. 3: .

5. (6 points) A voltage waveform as shown in Fig. 3(c) is applied to the circuit shown in Fig. 3(b).
- (a) (2 points) Compute the root mean square value of the input voltage.
- (b) (4 points) \*\*Assume that  $R \ll L$  (where  $\omega$  is the frequency of the input voltage) and  $RC \gg 1$ . Derive an expression for difference between the maximum and minimum value of inductor current in steady state. (Hint: try to relate the significance of time constant and how the assumptions simplify the analysis in steady state).